

PROJECT TITLE

SODA Challenge: Date Fruit Image Classification

A project submitted for the Machine learning for Data Science course (DS2401)

COURSE INSTRUCTOR

Machine learning

SUBMITTED BY:

Sarah MansourAlmalki446003470)

Lamar Mohammed Alghamdi (446002942)

Ritaj Mohammed Alhosawi (446007287)

Norah Salem Alqurbani (446002873)

DEPARTMENT OF DATA SCIENCE

COLLEGE OF COMPUTING

UMM AL-QURA UNIVERSITY

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ABSTRACT

This project aims to classify date fruit images into 13 categories using machine learning and deep learning techniques. The dataset was obtained from the SODA Challenge competition on Kaggle, containing 1,829 training images and 784 test images. Three models were built and compared: Random Forest, XGBoost, and a ResNet50 Transfer Learning model. The ResNet50 achieved the highest validation accuracy of 56.28%, outperforming the classical models. The results show that deep learning is more effective than classical machine learning for image classification, even with a relatively small dataset.

INTRODUCTION

Date fruits are one of the most important agricultural products in Saudi Arabia. Identifying different types manually is time-consuming and difficult, especially at a large scale. This project addresses this problem by building an automatic image classification system capable of distinguishing between 13 types of date fruits based on their visual appearance.

The main objective is to correctly classify date fruit images using both classical machine learning and deep learning approaches. The key research question is: which model performs best for this task? Since the input data consists of raw images, a deep learning approach was the most suitable choice, as it can automatically learn visual patterns such as color, shape, and texture directly from pixels.

METHODOLGY

DataSource&Description

The dataset was obtained from the SODA Challenge competition on Kaggle:

https://www.kaggle.com/competitions/soda-challenge

It contains 1,829 training samples and 784 test samples. Each entry includes two columns: id, which is the unique image identifier, and label, which represents the date fruit type as an integer from 1 to 13. The dataset contained no missing values and no duplicate rows. The classes were well balanced, with approximately 141 samples per class, as shown below:

Figure 1: Histogram of Label Distribution



Figure 2: Sample Training Images After Preprocessing

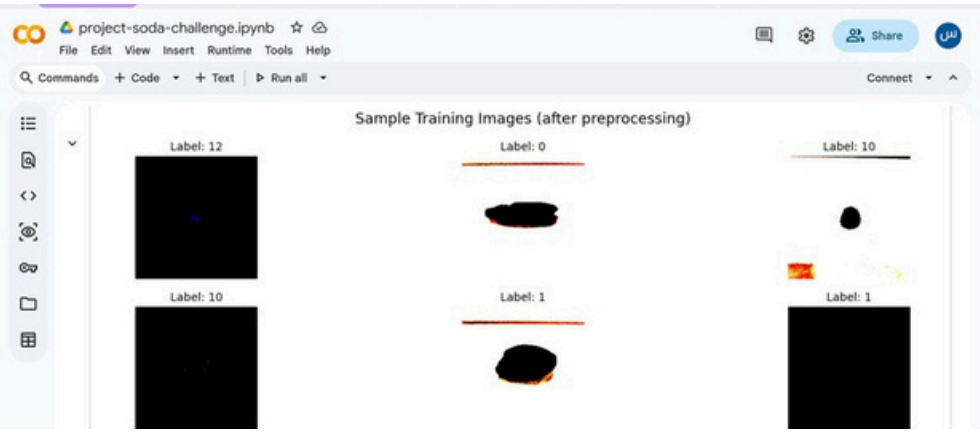




Table 1: Sample of Training Dataset

Shape : (1829 , 2)

	ID	label
0	1	13
1	2	1
2	3	11
3	4	11
4	5	2

Data Preprocessing:

Prior to model training, several preprocessing steps were applied to ensure data quality, improve model performance, and maintain consistency across the dataset. The dataset was first examined for potential issues such as missing values and duplicate entries. The inspection confirmed that the dataset contained no missing or duplicated samples, so no additional cleaning procedures were required.

Since the project involves image classification, all images were processed before training. Images were converted from the OpenCV BGR format to RGB to ensure correct color representation. They were then resized to 224 × 224 pixels, which is the required input size for the ResNet50 architecture. Standardizing image dimensions ensures consistency across samples and reduces computational complexity.

To enhance numerical stability and support efficient model convergence, pixel values were normalized from the original range of [0,255] to [0,1]. This normalization step helps the learning algorithm process the data more effectively.

The dataset was then split into training and validation sets using an 80:20 ratio. A stratified sampling strategy was used to preserve the original class distribution in both subsets, ensuring that validation performance accurately reflects the model's behavior across all classes.

To improve generalization and reduce overfitting, data augmentation techniques were applied during training. These included random rotations, zooming, horizontal flips, and width/height shifts. Augmentation increases the diversity of training samples and helps the model learn more robust visual features.

For the Random Forest and Gradient Boosting models, deep image features were extracted using a pretrained ResNet50 feature extractor. Before extraction, images were preprocessed using the ResNet50 preprocess_input() function to ensure compatibility with the pretrained network. This allowed traditional machine learning models to benefit from high-level visual representations.

Feature Extraction:

Since traditional machine learning algorithms cannot efficiently process raw image data, a feature extraction step was performed prior to training the Random Forest and Gradient Boosting models. A pretrained ResNet50 network was used to extract high-level visual representations from the date fruit images.

Before extraction, all images were resized to 224 × 224 pixels and preprocessed using the ResNet50 preprocess_input() function to match the model's input requirements. The final classification layer of ResNet50 was removed, and the output of the convolutional layers was used as a feature vector for each image.

These extracted features capture important visual characteristics such as texture, shape, color patterns, and structural details. The resulting feature vectors were then used as inputs for the Random Forest and Gradient Boosting classifiers.

Using transfer learning for feature extraction eliminates the need for manual feature engineering and provides more informative representations compared to handcrafted features, ultimately improving classification performance.

Algorithm/Model Selection:

Threedifferentmodelswere implemented and compared for the date fruit image classification task: Random Forest, XGBoost, and ResNet50 Transfer Learning.
Random Forest was selected as a baseline machine learning model because it is robust, easy to implement, and performs well on high-dimensional feature representations. Since Random Forest cannot process raw images directly, image features were first extracted using a pretrained ResNet50 network and then used as input for classification.
XGBoost was selected as an advanced ensemble learning method that builds decision trees sequentially to improve classification performance by correcting previous prediction errors. Similar to Random Forest, it was trained using image features extracted from a pretrained ResNet50 model.
ResNet50 Transfer Learning was selected because it is specifically designed for image recognition tasks. A pretrained ResNet50 model with ImageNet weights was used as the backbone network, allowing the model to leverage previously learned visual features and adapt them to the date fruit classification task.

The final model was selected by comparing the performance of all implemented models using evaluation metrics including Accuracy, Classification Report, and Confusion Matrix. The model that achieved the best overall classification performance was considered the final model.

Training Procedure:

The datasetwas dividedinto training and validation sets using an 80% training and 20% validation split. Stratified sampling was applied to preserve the original class distribution in both datasets and ensure a fair evaluation process.
For the Random Forest and XGBoost models, image features were extracted using a pretrained ResNet50 feature extractor. These extracted features were then used to train the machine learning classifiers.
For the ResNet50 Transfer Learning model, the original images were used directly as input. The pretrained ResNet50 backbone was loaded with ImageNet weights, and additional classification layers were added to perform the final prediction task.

The convolutional layers of the pretrained network were frozen during training, while the newly added classification layers were trained on the date fruit dataset.

Cross-validation was not applied in this project. Instead, a stratified 80/20 train-validation split was used as a practical validation strategy. This approach reduced computational cost while maintaining reliable performance evaluation and preserving class balance across the datasets.

Evaluation Strategy:

The performance of the proposed image classification models was evaluated using a stratified 80/20 train-validation split to ensure that all date fruit classes were proportionally represented in both datasets. Three different models were trained and compared: Random Forest, XGBoost, and ResNet50 Transfer Learning.

Model performance was primarily assessed using Accuracy, Classification Report, and Confusion Matrix. Accuracy was selected because the task is a multi-class image classification problem and provides an overall measure of correctly classified samples. The Classification Report was used to evaluate precision, recall, and F1-score for each class, providing a more detailed assessment of classification performance. The Confusion Matrix was used to visualize classification errors and identify which date fruit classes were frequently confused with each other.

To improve model generalization, image preprocessing techniques such as image resizing, normalization using preprocess_input(), and data augmentation were applied. In addition, EarlyStopping was used during ResNet50 Transfer Learning training to reduce overfitting by stopping training when validation performance stopped improving. Due to computational limitations and training time constraints, k-fold cross-validation was not applied. Instead, a stratified train-validation split was used to provide a reliable estimate of model performance.

The implementation was carried out using Python. Major libraries included TensorFlow and Keras for deep learning, Scikit-learn for Random Forest and evaluation metrics, XGBoost for gradient-boosted decision trees, OpenCV for image preprocessing, NumPy and Pandas for data handling, and Matplotlib and Seaborn for visualization.

RESULTS AND DISCUSSION

Model Performance Comparison

Three classification models were implemented and evaluated for date fruit image classification: Random Forest, XGBoost, and ResNet50 Transfer Learning.

Model	Input	Accu racy (%)
ResNet50 Transfer Learning	Raw images (224×224×3)	56.28
XGBoost	ResNet50 extracted features (2048-dim)	50.00
Random Forest	ResNet50 extracted features (2048-dim)	46.99

The results indicate that the ResNet50 Transfer Learning model achieved the highest validation accuracy of 56.28%, outperforming both XGBoost (50.00%) and Random Forest (46.99%).

Despite these limitations, the results demonstrate that deep learning approaches—particularly transfer learning—still outperform traditional machine learning methods for image classification tasks.

Why ResNet50 Transfer Learning Performed Best

trained directly on raw image pixels using pretrained ImageNet weights. This approach allowed the model to adapt its learned representations to the date fruit classification task. Unlike classical machine learning models such as Random Forest and XGBoost, which rely on fixed pre-extracted ImageNet features, the ResNet50 Transfer Learning model was able to leverage existing visual knowledge and adjust it to capture subtle differences between date fruit categories.

Dropout (0.5) was applied during training to reduce overfitting, and EarlyStopping was used to stop training once validation performance stopped improving. Although the overall accuracy remained relatively low due to the small dataset size and the high visual similarity between classes,the ResNet50 Transfer Learning model still achieved the best performance among all evaluated models.

Why Random Forest and XGBoost Performed Lower

feature vectors extracted from a pretrained ResNet50 model. These features were originally learned from the ImageNet dataset and may not fully represent the fine-grained visual characteristics needed to distinguish between similar date fruit categories. Because both models rely on fixed, pre-extracted features, they are unable to learn new visual patterns directly from the raw images.

As a result, their performance is limited by the quality and relevance of the extracted features. Unlike the ResNet50 Transfer Learning model, which leverages pretrained ImageNet weights and adapts them to the date fruit classification task, Random Forest and XGBoost cannot update or refine the underlying visual features, leading to lower classification accuracy.

Model Generalization

To assess model generalization, the dataset was divided using a stratified 80/20 train-validation split, ensuring that all date fruit classes were proportionally represented in both subsets.

Training and validation performance were monitored throughout the process. EarlyStopping was used to reduce the risk of overfitting by stopping training when validation accuracy no longer improved.

The ResNet50 Transfer Learning model benefited from pretrained ImageNet weights, which helped improve generalization despite the limited dataset size.

Evaluation Metrics

Thefollowingevaluation metrics were used:

- Accuracy
- Classification Report (Precision, Recall, and F1-Score)
- Confusion Matrix

Accuracy was selected as the primary metric because the task is a multi-class image classification problem. Precision, Recall, and F1-Score provided additional insight into the classification performance for each class. The Confusion Matrix was used to visualize prediction errors and identify frequently confused classes.

Libraries and Tools

The project was implemented in Python using:

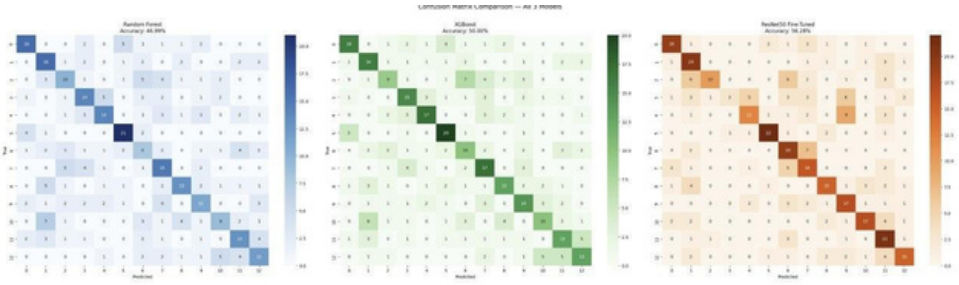
- TensorFlow / Keras
- Scikit-learn
- XGBoost
- NumPy
- Pandas
- Matplotlib
- Seaborn

Limitations

Several limitations may have affected the final performance of the models:

- The dataset contains only **1,829 training images**, which limits the model's ability to generalize and makes deep learning training more challenging.
- Some date fruit classes exhibit **high visual similarity**, making it difficult for the models to distinguish between them.
- **K-fold cross-validation** was not applied due to computational constraints, which may have reduced the reliability of the validation accuracy.
- More advanced **data augmentation**, **transfer learning strategies**, and **hyperparameter optimization** could potentially improve performance.
- The relatively low validation accuracy, especially for the deep learning model, is expected given the **small dataset size** and the **complexity of the classification task**.

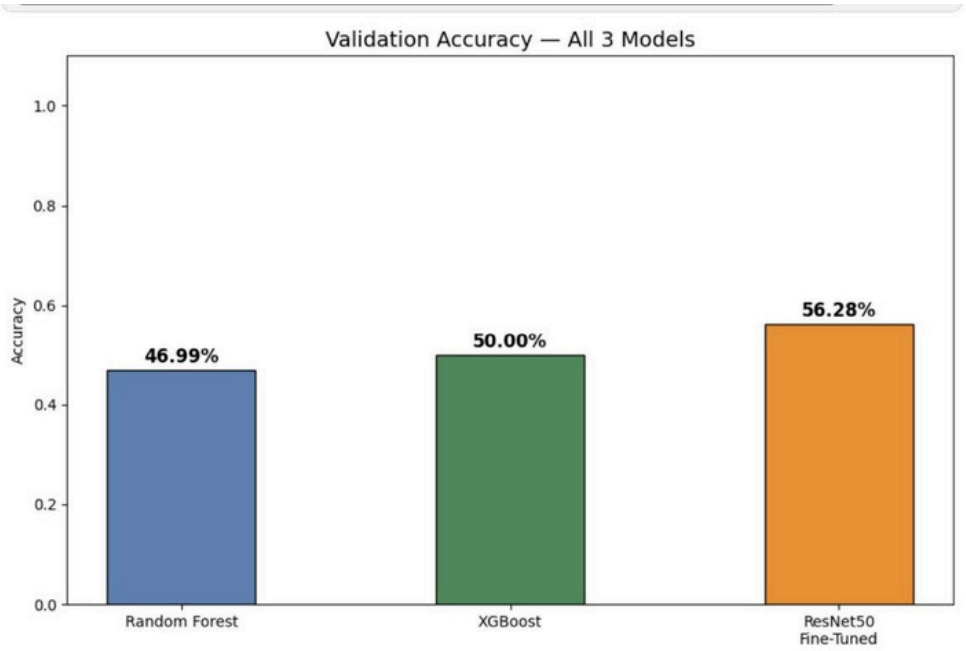
Figure 4: Confusion Matrix (Random Forest,XGBoost ,ResNet50)



The figure compares the confusion matrices of the three models: Random Forest, XGBoost, and ResNet50 Transfer Learning. Correct classifications are represented by the diagonal values, while off-diagonal values indicate misclassifications.

The results show that ResNet50 Transfer Learning achieved the highest classification accuracy (56.28%), followed by XGBoost (50.00%) and Random Forest (46.99%). The ResNet50 confusion matrix contains stronger diagonal values and fewer classification errors, indicating a better ability to distinguish between date fruit classes. However, some confusion between visually similar classes is still observed, which may be attributed to the similarity of date varieties and the limited dataset size.

Figure 5: Model Accuracy Comparison



The figure compares the validation accuracy of the three models. ResNet50 Transfer Learning achieved the highest accuracy (56.28%), followed by XGBoost (50.00%) and Random Forest (46.99%). These results indicate that the deep learning model was more effective in learning visual features from date fruit images and achieved the best overall performance.

CONCLUSION

The objective of this project was to classify date fruit images into their corresponding categories using machine learning and deep learning techniques. Three models

were implemented and compared: Random Forest, XGBoost, and ResNet50 Transfer Learning. Image preprocessing techniques such as image resizing, normalization, and feature extraction were applied prior to training. Random Forest and XGBoost were trained using features extracted from a pretrained ResNet50 network, while the ResNet50 Transfer Learning model was trained directly on image .data using pretrained ImageNet weights and custom classification layers

The experimental results showed that the ResNet50 Transfer Learning model achieved the highest validation accuracy of 56.28%, outperforming XGBoost (50.00%) and Random Forest (46.99%). Although the overall accuracy remained relatively moderate, this outcome is expected due to the limited dataset size and the high visual similarity among several date fruit classes. The results indicate that transfer learning and deep learning approaches can learn more discriminative visual representations than traditional machine learning models that rely on fixed extracted .features

Despite achieving the best performance among the evaluated models, classification errors were still observed due to dataset limitations and the complexity of distinguishing between visually similar date varieties. Future work could focus on collecting additional training data, performing hyperparameter optimization, exploring more advanced deep learning architectures, and applying fine-tuning to selected .layers of the pretrained network to further improve classification performance

Overall, this project demonstrated the effectiveness of machine learning and transfer learning techniques for image classification tasks. The developed approach has potential applications in agriculture, quality control, and automated date fruit recognition systems, contributing to more efficient and accurate classification .processes in real-world environments

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REFERENCES

- Support your analysis and methodology with at least two academic references.
- You may use journal articles, conference papers, textbooks, or reliable online resources (e.g., scikit-learn documentation).
- Your references must be cited properly within the report and listed in APA style at the end. cite any data sources. This means including (Author, Year) in your sentences where the references are introduced, and then listing the full reference in your References section.

- He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep residual learning for image recognition*. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770–778.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). *Scikit-learn: Machine learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
- Kaggle. (2026). *SODA Challenge Dataset*. Retrieved from Kaggle Competition Dataset.

PARTICIPATION, EXPERIENCES AND SKILLS ACQUIRED BY THE TEAM MEMBERS :

Students name	**Contribution to the project**	**Experiences and Skills Acquired**
Sarah Mansour Almalki	I worked on data loading, preprocessing, handling missing values, normalizing images, and evaluating the three classification models.	Ilearned how to preprocess image data, use ResNet50 for feature extraction, and evaluate models using accuracy and confusion matrix
Ritaj Mohammed Alhosawi	I handled the image preprocessing steps, including loading, resizing, normalizing images, fixing label ranges, performing a stratified train–validation split, and extracting features using the pretrained ResNet50 model.	I gained experience in image preprocessing, label correction, feature extraction using ResNet50, and improved my skills in Python, OpenCV, and TensorFlow.
Lamar Mohammed Alghamdi	Implemented and trained the machine learning and deep learning models used in the project, including Random Forest, XGBoost, and ResNet50 Transfer Learning. Configured the model architectures and training parameters to support the date fruit image classification task	Developed skills in machine learning and deep learning model implementation, model training, transfer learning using ResNet50, hyperparameter configuration, and image classification using Python, Scikit-learn, TensorFlow, and Keras
Norah Salem Alqurbani	**Conducted the evaluation and comparison of the three models (Random Forest, XGBoost, ResNet50 Transfer Learning)** **Analyzed accuracy, classification reports, and confusion matrices.** • **Summarized model performance and wrote the final model analysis section.**	**Gained experience in evaluating machine learning and deep learning models.** • **Improved skills in interpreting accuracy metrics and confusion matrices.** • **Developed the ability to compare models and write analytical performance summaries**